

Adiba Shaira

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Summary

Ph.D. candidate in Computer Science specializing in wearable health sensing, multimodal time-series machine learning, and Large Language Model (LLM) applications in digital health. Hands-on experience building end-to-end pipelines from raw physiological signals to interactive health insights.

Education

Ph.D. in Computer Science, Stony Brook University (Candidate) Aug 2023 – Aug 2027 (expected)
M.S. in Computer Science, Stony Brook University, GPA: 3.95/4.00 Dec 2025

- PiCASSo Lab; advised by Prof. Shubham Jain
- **Research Areas:** Health Sensing, Wearable Computing, Time-Series Modeling, Digital Phenotyping, Machine Learning, Large Language Models (LLMs), Explainable Machine Learning

B.Sc. in Computer Science and Engineering, Bangladesh University of Engineering and Technology May 2022

Experience

Research Assistant, Stony Brook University May 2024 – Present

- Built LiftSafe, a machine learning system that predicts unsafe strength-training repetitions one repetition ahead from a single arm-worn Inertial Measurement Unit (IMU), reaching 87% balanced accuracy in a 24-participant study with EMG ground truth; published at IEEE/ACM CHASE 2026.
- Built CareQA, a multi-agent LLM system for context-aware sensemaking that helps caregivers interpret their autistic child's sleep, activity, and physiological signals by grounding answers to natural-language questions in the child's own passively sensed data and clinical context from the [Simons Sleep Project](#).
- Designing passive smartphone-based sleep sensing for PTSD monitoring in a World Trade Center 9/11 first-responder cohort: extracting sleep metrics from accelerometer and screen-state data and correlating them with clinical PCL scores via longitudinal statistical modeling.

Teaching Assistant, Stony Brook University: CSE 373, Analysis of Algorithms Aug 2023 – May 2024

Lecturer, United International University, Dhaka, Bangladesh May 2022 – July 2023

- Taught Human-Computer Interaction (HCI), Analysis of Algorithms, and Computer Networks.

Publications

CareQA: Multi-Agent Sensemaking to Answer Caregiver Queries from Passive Sensing Data in Autism Care

Adiba Shaira, Michelle S. Ballan, Shubham Jain. *Under submission at IEEE PerCom 2027.*

LiftSafe: Predicting Unsafe Repetitions in Strength Training using a Single Wearable Sensor

Adiba Shaira, Hunter J. Bennett, Shubham Jain. *IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (CHASE)*, 2026.

Gene Tree Parsimony in the Presence of Gene Duplication, Loss, and Incomplete Lineage Sorting

Prottoy Saha, Md. Shamiul Islam, Tasnim Rahman, **Adiba Shaira**, Kazi Noshin, Rezwana Reaz, Md. Shamsuzzoha Bayzid. *RECOMB International Workshop on Comparative Genomics*, Springer Nature, pp. 110–128, 2024.

Other Projects

Using a Single Device to Predict Sleep Efficiency and Next-Day's Mood 2024

Modeled next-day sleep efficiency and mood from single-wearable time-series data in the GLOBEM dataset. *Key Techniques:* Python, Pandas, NumPy, Scikit-learn, LSTM, Autoencoder models (PyTorch/TensorFlow)

Global Disaster Visualizer 2023

Interactive web-based visualization of global disaster data. *Key Techniques:* D3.js, JavaScript, React, Python

Awards

NSF Student Travel Award, IEEE/ACM CHASE 2026

Distinguished Travel Award, Graduate Student Organization, Stony Brook University 2026

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, SHAP, Large Language Models (LLMs), Multi-Agent LLM Systems, Retrieval-Augmented Generation (RAG)

Sensing & Data: IMU data processing, physiological signals (PPG, EDA, EEG), OpenCV, MATLAB, Pandas, NumPy

Tools: Git, Jupyter, Wireshark, Django, Bootstrap, MySQL, MongoDB